



BUYERS' GUIDE

ESTYLE

Choosing BLOOD SUGAR MONITORS



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Sweet, delicious, mouth-watering and zero calories! Double trouble as global blood sugar levels in humans are shooting phenomenally, along with the numbers appearing on sugar monitors. There are many sugar-monitoring devices available in the market but most of us realise the need of having one only after it is too late or our dietician suggests urgency. It is time we know that these devices have more than what meets the eye, and the guide to knowing what these are made of starts here.

Measuring glycaemia is not scary

Let us understand how a blood sugar monitor works. It lets you measure your blood sugar level at any given point of time. Primarily, the sugar profile, or glycaemia, portrays a good picture of what is happening with respect to home blood glucose monitoring and how sugar is playing a vital role in your system. This helps dieticians and doctors administer medication to alleviate imbalance in blood sugar levels.

You are required to prick your finger or any fleshy part of your hand so a small blood drop appears. This is touched with a disposable test strip fitted in the meter. Sugar count instantly gets displayed on the meter's LCD or OLED screen. The lancet (pin prick) that comes with the kit pokes the body surface (usually fingertip) in such a way that it does not create a deep cut.

Some people feel taking blood from

fingertips is painful. Alternate-site testing comes to their rescue as most devices can take blood sample from places other than fingertips as well. It may relieve you of the pain but accuracy in reading may get affected.

Testing blood glucose



A good blood sugar monitor is expected to show results in less than five seconds. Accuracy is important, too. For example, if you have hypoglycaemia, that is, low blood sugar, you may need to consume sugar or glucose immediately in order to avoid dizziness or related health problems.

Palm-sized chemical labs

What is the technology behind these little wonders of technology? Read on.

Active chemical ingredients. Biosensors are at the heart of every blood sugar monitoring device. These are based on a thin layer of glucose-oxidase (GOx) on an oxygen electrode. Measurement of sugar is based on the amount of oxygen consumed by GOx while reacting with glucose on the sensor strip.

Biosensor technology. Once the blood passes through the GOx electrode, a charge is generated due to re-oxidation reaction at the electrode in presence of a mediator reagent. This charge is directly proportional to the glucose present in the blood.

Then there are amperometric and coulometric techniques, which measure the electric current generated at a certain point and variable point in time, respectively, during glucose reaction.

What you should look for

There are scores of interesting bits you should look for before selecting a blood glucose monitor. These are:

More memory. Memory size in such devices decides the number of records these can store. Some companies claim that they give additional memory of a few hundred megabytes to add more personal data.

Quicker results. Optimally, results on these devices are expected to show up in less than five seconds. If you are at a higher risk and require faster attention in emergency situations, you can rely on its swiftness; so that effects of dwindling